CLINICAL-ANCHORED MULTI-OMICS GRAPH LEARNING FOR BREAST CANCER SURVIVAL: AN AUDITED METABRIC EVALUATION

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ABSTRACT

Multi-omics survival models require patient-level matching and evaluation that distinguishes representation learning from information leakage. We adapt a preliminary four-view canonical-correlation graph model to METABRIC, using mRNA, promoter methylation, copy-number alterations and baseline clinical covariates. A regularized generalized canonical correlation analysis (GCCA) uses one shared latent target with an explicit clinical-view weight; a static graph combines the available projected views and predicts overall survival. We evaluate 1,416 patients with 830 deaths using five patient-level held-out folds, validation-only model selection and three seeds, comparing fifteen main configurations including survival-specific and adapted multi-omics baselines. The shared-anchor graph achieved a mean C-index of 0.607 (conditional 95% CI 0.587–0.625), compared with 0.656 for Structured clinical Cox. The difference was -0.049. All fifteen nonempty view subsets, five component ablations and source-group holdouts are evaluated. The fusion configuration did not outperform the strongest evaluated baseline. The experiment supports a reproducible evaluation framework rather than a claim of superior prognosis prediction. There are no repeated visits, paired images or independent external cohort.

***Index Terms*—** Multi-omics, breast cancer, survival, GCCA, graph learning

1. INTRODUCTION

Breast cancer prognosis depends on clinical disease burden and molecular heterogeneity. METABRIC provides expression, copy-number and clinical annotations for breast tumours, with additional molecular profiles available for partially overlapping samples [1, 2]. Combining these assays may improve individual risk estimates, but increasing model complexity does not ensure that the additional information generalizes. Matched patient identifiers, genuinely baseline covariates and a consistent evaluation target are prerequisites for any meaningful comparison.

Graph-based multi-omics methods offer a natural way to model relationships between patients. MOGONET uses view-specific graph encoders and cross-view classification-label interactions [3]; IntegrAO learns aligned representations from partially overlapping molecular graphs [4]. These contributions do not, by themselves, establish an advantage for a new Cox survival model. Comparisons must distinguish original task definitions from downstream adapters, and must keep test patients outside representation fitting and graph diffusion. Survival-specific baselines, including DeepSurv and MultiSurv, provide complementary nonlinear approaches [5, 6].

The supplied preliminary design aligned three molecular views independently to templated clinical text and then averaged the clinical projections. Independently fitted canonical coordinates can differ in sign and orientation; arithmetic averaging does not resolve this ambiguity. We therefore evaluate a shared-target regularized GCCA formulation, alongside that preliminary pairwise construction. GCCA is an established estimator [7]; the clinical weight and its combination with missing-aware inductive graph prediction are implementation choices to test, not claims of a new CCA theory.

Our contribution is an auditable, static survival-learning evaluation: (i) identifier-based alignment of four METABRIC views; (ii) shared versus pairwise clinical alignment and a training-reference graph; and (iii) matched comparisons, component ablations and missing-view stress tests. We explicitly report negative findings and adapted baseline architectures. This work is a multi-omics prognosis study, not an imaging-fusion validation or a temporal disease-progression study.

2. METHODS

2.1. Patients, views and preprocessing

From 2,509 clinical records, 1,980 had positive overall-survival time and a recognized event/censoring code. Intersecting the three assay sample sets with primary tumours and valid survival yielded 1,416 unique patients, including 830 deaths. Before training, all views were ordered by the same verified patient/sample mapping. Duplicate gene symbols were retained as distinct probe/promoter rows, not collapsed into an unsupported biological identity.

Expression measurements were already $\log_2$ intensities and were not logged again. Promoter methylation beta values were transformed to $\log_2[\beta/(1-\beta)]$ after clipping to $[10^{-5}, 1-10^{-5}]$; discrete copy-number calls were retained. Within each training fold, assay features with over 20% missing observations were excluded, 500 or 1,000 high-variance features were selected, missing entries were median-filled, and features were standardized and clipped at five training standard deviations. No feature selection, vocabulary fitting or scaling used validation/test patients.

The clinical view contained age, sex, stage, grade, ER, PR and HER2 status, tumour size, histological type and menopausal state. Stage was missing in 26.7% of the included patients. Outcome, recurrence and treatment fields were excluded. Fixed field–value strings were encoded with 1–2-gram TF-IDF and training-only SVD (at most 128 dimensions). These are templated structured covariates, not free-text notes or an additional information source. A structured numerical and categorical encoder was evaluated as an ablation.

2.2. Shared alignment and static graph

Let X_v be centered training features for view v, and let a_4 denote the clinical-view weight; other a_v equal one. We solve the regularized MAXVAR GCCA problem

$$\min_{G,\{W_v\}} \sum_{v=1}^{4} a_v \big(\|X_v W_v - G\|_F^2 + n\lambda \|W_v\|_F^2\big), \quad G^\top G = nI. \tag{1}$$

The leading eigenvectors of $\sum_v a_v H_v$ give G, where $H_v = X_v(X_v^\top X_v + n\lambda I)^{-1} X_v^\top$; primal or dual solves are chosen by matrix size. The projections are then fitted to this single target. The comparator fits three independent molecular–clinical ridge CCAs and averages the clinical projections as in the preliminary design.

For availability indicators m_{iv}, graph features are $\bar{z}_i = (\sum_v m_{iv} z_{iv})/(\sum_v m_{iv})$. A symmetric binary cosine k-nearest-neighbour graph is normalized after adding self-loops. Two residual GCN layers with LayerNorm and dropout produce the graph embedding. The Cox head receives this embedding, each masked view projection and the four availability indicators. Independent view masking during training is conditioned to leave at least one view; an otherwise empty draw retains mRNA.

The training graph contains only training patients. At inference, each query receives messages from its nearest training nodes and itself; training-node states and degrees are frozen. Queries have no edges to each other or back to training nodes. Thus a prediction does not depend on other patients in its test batch. No visit times, survival times or event indicators enter the graph.

2.3. Learning and validation

The neural Cox head minimizes Breslow negative partial log likelihood

$$\mathcal{L} = -\frac{1}{\sum_i \delta_i} \sum_{i:\delta_i=1} \left[r_i - \log \sum_{j:t_j \geq t_i} \exp(r_j) \right]. \tag{2}$$

Five event-stratified patient folds were fixed with seed 42. Each development set reserved 20% for validation (approximately 64/16/20% overall). Up to twelve configurations per method were evaluated using validation C-index; the winner was rerun with seeds 42, 2026 and 3407. Neural training used AdamW, weight decay 10^{-3}, at most 300 epochs and patience 30. Ablations reused the selected proposed-model settings. Full search and selected values are reported in the separate appendix and machine-readable parameter tables.

3. EXPERIMENTS AND RESULTS

3.1. Matched baselines and discrimination

The main comparison included five unimodal regularized Cox models (three molecular views and two clinical encodings), concatenation Cox, a survival forest, a DeepSurv-style Cox MLP, pairwise CCA–Cox, equal-weight GCCA–Cox, the preliminary pairwise graph, a scaled MultiSurv-style discrete-time model, a MOGONET-style multi-graph Cox adapter, official IntegrAO embeddings with an inductive Cox adapter, and the proposed configuration. The independently reimplemented/scaled baselines are labeled as such; they are not exact reproductions of every original architecture. IntegrAO representation settings were fixed, with validation selection only for its Cox regularization. All other model-selection details are in the appendix.

Table 1 reports the mean test C-index over five folds and three seeds. Patient bootstrap intervals (1,000 resamples) preserve held-out fold membership and use the same resampled patients for method comparisons. They condition on the fitted models and do not include retraining or new-center uncertainty.

Table 1. Held-out overall-survival discrimination.

Method	C-index	95% CI
Structured clinical Cox	0.656	0.634–0.675
Survival forest	0.620	0.600–0.639
mRNA Cox	0.619	0.599–0.640
Pairwise CCA + Cox	0.619	0.599–0.639
Templated clinical Cox	0.618	0.596–0.638
DeepSurv-style	0.607	0.587–0.626
Shared-anchor graph	0.607	0.587–0.625
MOGONET-style + Cox	0.605	0.585–0.625
Concatenation Cox	0.604	0.584–0.625
GCCA + Cox	0.598	0.576–0.621
Pairwise CCA + GCN	0.591	0.575–0.609
MultiSurv-style (scaled)	0.586	0.567–0.605
IntegrAO + Cox adapter	0.576	0.557–0.597
Methylation Cox	0.544	0.523–0.567
CNA Cox	0.515	0.492–0.538

The shared-anchor graph achieved a mean C-index of 0.607 (conditional 95% CI 0.587–0.625), compared with 0.656 for Structured clinical Cox. The difference was -0.049. The paired bootstrap difference against the strongest baseline had 95% interval -0.075 to -0.025 (Holm-adjusted $p = 0.038$). The structured clinical-only model reached 0.656, making it an essential control for any claimed benefit from expensive molecular views. IPCW concordance, dynamic AUC and Brier scores at one, three and five years are reported in the appendix. Censoring estimates and Cox baseline hazards were fitted on training data. Time-dependent evaluation administratively censored held-out observations beyond training follow-up support.

3.2. Missing views, ablations and source groups

All fifteen nonempty availability patterns were evaluated without refitting the model on the test patients. Table 2 summarizes clinically interpretable single-view omissions; all combinations and random missingness at 10, 30 and 50% appear in the appendix. Missingness was simulated independently and should not be interpreted as a clinical missing-not-at-random process.

Table 2. Proposed model under view omission.

Available input	C-index
Without mRNA	0.579
Without methylation	0.604
Without CNA	0.617
Without clinical	0.592
All four views	0.607

Table 3. Component ablations (fixed selected settings).

Configuration	C-index
Structured clinical	0.655
Without raw residual	0.609
Shared-anchor graph	0.607
Without graph	0.602
Equal view weights	0.601
Without mask training	0.597

The structured-clinical graph variant reached 0.655, close to the structured clinical Cox point estimate of 0.656. This comparison does not establish added value from graph modeling. The equal-weight ablation removes additional clinical weighting, not the clinical view itself. If the proposed search selects weight one, that fold's anchor ablation is identical by construction. This possibility is retained rather than forcing an artificial distinction. The structured-clinical comparison tests representation, since both encoders receive the same fields.

Across five source-group holdouts, the proposed model had mean C-indices ranging from 0.562 to 0.658. These groups are METABRIC source labels and are not an independent external breast-cancer cohort. All group-specific models were fitted without the held-out group; results include the clinical, concatenation, GCCA, original-graph, proposed and IntegrAO controls.

4. DISCUSSION

The fusion configuration did not outperform the strongest evaluated baseline. The experiment supports a reproducible evaluation framework rather than a claim of superior prognosis prediction. The clinical-only control and the explicitly adapted modern baselines are particularly informative: a flexible graph may preserve correlated molecular structure without improving the ranking of survival times. Added model capacity, assay noise, limited sample size and the objective mismatch between unsupervised correlation and censored outcome prediction are plausible explanations to investigate, not mechanisms established here.

The shared GCCA target resolves a coordinate consistency problem in the preliminary pairwise design, but correcting that construction is not proof of prognostic superiority. Likewise, accurate operation under a specified view-availability pattern does not establish calibrated uncertainty or an error-control guarantee. The supplied rejection and Learn Then Test papers concern different prediction losses; we do not transfer their guarantees to censored survival outcomes without a corresponding theory and experiment.

Several limitations bound the interpretation. First, selecting complete assays reduces the available survival cohort from 1,980 to 1,416 patients and may introduce selection bias. Second, validation and test patients originate from one study, and the additional source-group holdouts do not replace external validation. Third, clinical strings are an encoding of measured fields; they cannot establish a language-model or clinical-note benefit. Fourth, there are no repeated visits and no matched patient images. Accordingly, the temporal and imaging extensions in the preliminary proposal remain unevaluated.

The survival head makes a proportional-hazards modeling assumption, and Breslow-derived survival probabilities may be miscalibrated even when ranking performance is acceptable. Calibration and out-of-fold Kaplan–Meier groups are therefore reported separately. Bootstrap intervals are conditional on this experiment, while subgroup analyses and simulated missingness are exploratory. Architecture-scaled baseline reimplementations are useful controls but do not establish a complete state-of-the-art ranking.

5. CONCLUSION

We provide patient-aligned four-view METABRIC preprocessing, shared and pairwise canonical-correlation models, an inductive static graph, and a reproducible survival benchmark with missing-view evaluation. The results support only the performance comparisons measured here. Independent external validation, genuinely longitudinal assays and paired imaging would be required before extending the scientific or clinical claims.

6. COMPLIANCE WITH ETHICAL STANDARDS

This is a secondary analysis of publicly distributed, de-identified METABRIC research data. No new patient data were collected. The responsible authors must confirm their institution's review/waiver determination before any submission; no approval identifier or exemption is invented here.

7. ACKNOWLEDGMENTS AND DISCLOSURE

This preliminary draft requires author, affiliation, funding and conflict-of-interest confirmation. Codex assisted with implementation, experiment execution and draft text throughout the methods/results manuscript; accountable human authors must verify the analyses, references and conclusions before submission. Code and aggregate tables are prepared for a public repository; patient-level clinical data and predictions are excluded. The separate appendix is an author-review supplement, not an assumed permitted extra technical submission page.

8. REFERENCES

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